Supplementary Material

How Much Reported Breast-Cancer AI Accuracy Survives Honest Validation? An Optimism-Ladder Audit of Metastasis Classifiers and an Audit-Ready Digital Twin

All numbers below are produced by the two released notebooks: CaseStudy1_MetastasisLadder.ipynb (public GEO data; seed 20260806) and PECT_DigitalTwin_Audit_v3.ipynb (digitised PECT data released as CSV; independent seeds per analysis).

S1. Case study 1: patient flow (TRIPOD+AI)

Table S1. Patient flow. GSE2603: 121 arrays, 22 cell-line samples removed. Five-year distant metastasis: event by 60 months = 1; follow-up of at least 60 months = 0; censored earlier = excluded (GSE7390, GSE11121: t.dmfs/e.dmfs; GSE2603: authors' binary 5-year field). Bone relapse is recorded only in GSE2034 and GSE2603.

Cohort	Primary-tumour arrays	5-yr DM evaluable	5-yr DM events	Excluded (censored < 5 y)	Bone evaluable	Bone events	Size and grade complete
GSE2034	286	0	0		286	69	0
GSE11121	200	182	28	18	0	0	200
GSE2603	99	68	22	31	82	14	0
GSE7390	198	190	36	8	0	0	196

S2. Case study 1: preprocessing, models and protocols

Genes: GPL96 probes annotated to exactly one gene symbol; for each gene the probe with the highest mean log2 expression over all 783 arrays (label-free) → 12,502 genes. GSE2034, GSE11121 and GSE2603 series matrices are MAS5 intensities (log2 of max(x,1)); GSE7390 is supplied on the log2 scale.

Fold-internal preprocessing: quantile normalisation to the mean sorted training distribution (each sample mapped individually); parametric empirical-Bayes ComBat fitted on training cohorts; held-out samples from a cohort present in training are adjusted with that cohort's estimates, an unseen cohort is not batch-adjusted; genes standardised with training means and SDs; 50 genes selected by |AUC − 0.5| on training samples.

Models (scikit-learn 1.8): penalised logistic regression (L2, C = 0.1, balanced class weights); RBF SVM (C = 1, gamma = scale, Platt probabilities, balanced); random forest (400 trees, min leaf 3, sqrt features, balanced); gradient boosting (500 trees, learning rate 0.03, depth 3, subsample 0.8, max features 0.6, balanced sample weights; replaces XGBoost); histogram gradient boosting (500 iterations, learning rate 0.03, 7 leaves, depth 3, L2 = 2, balanced; replaces LightGBM); feed-forward network (64–32, alpha 0.01, early stopping).

Protocols: L1 leave-one-cohort-out; L2 4 × stratified 5-fold CV, preprocessing and selection inside folds; L3 as L2 with the 50 genes selected once on all samples; L4 20 stratified 80/20 splits with leaky selection, best of six models reported; L5 resubstitution. Mean AUC in the ladder = mean over the six families of (L1) the mean of cohort AUCs, (L2–L3) the mean over 20 folds, (L4) the best model per split averaged over splits, (L5) the training AUC. Sensitivity analyses for L1: each unseen test cohort centred and scaled per gene on its own unlabelled data; and quantile normalisation plus ComBat fitted on all 783 arrays before splitting.

Table S2. AUC by model family and rung.

Endpoint	Model	L1_strict	L1_test_self	L1_global	L2	L3	L4_single_model	L5

DM5	Penalised LR	0.635	0.635	0.635	0.663	0.741	0.715	0.852
DM5	SVM (RBF)	0.619	0.619	0.617	0.662	0.743	0.741	0.955
DM5	Random forest	0.657	0.622	0.618	0.682	0.730	0.727	1.000
DM5	Gradient boosting	0.628	0.578	0.577	0.639	0.696	0.664	1.000
DM5	Hist. gradient boosting	0.631	0.595	0.598	0.638	0.683	0.664	1.000
DM5	Neural network	0.633	0.621	0.613	0.632	0.618	0.619	0.601
DM5	L4 best of six (mean over splits)						0.757	
BONE	Penalised LR	0.588	0.597	0.594	0.670	0.799	0.813	0.907
BONE	SVM (RBF)	0.568	0.568	0.573	0.657	0.799	0.811	0.987
BONE	Random forest	0.562	0.555	0.553	0.652	0.804	0.820	1.000
BONE	Gradient boosting	0.585	0.579	0.576	0.658	0.795	0.804	1.000
BONE	Hist. gradient boosting	0.565	0.567	0.544	0.647	0.790	0.782	1.000
BONE	Neural network	0.577	0.601	0.604	0.623	0.744	0.727	0.740
BONE	L4 best of six (mean over splits)						0.834	

Table S3. Strict leave-one-cohort-out AUC per held-out cohort (DeLong 95% CI) and DerSimonian–Laird pooled estimate.

Endpoint	Model	GSE11121 AUC (95% CI)	GSE2603 AUC (95% CI)	GSE7390 AUC (95% CI)	Random effects (95% CI)	I^2 (%)	GSE2034 AUC (95% CI)
DM5	Penalised LR	0.671 (0.556–0.786)	0.619 (0.478–0.759)	0.615 (0.517–0.713)	0.633 (0.565–0.696)	0	
DM5	SVM (RBF)	0.671 (0.559–0.784)	0.549 (0.406–0.693)	0.635 (0.531–0.740)	0.626 (0.556–0.692)	0	
DM5	Random forest	0.696 (0.591–0.801)	0.605 (0.467–0.742)	0.669 (0.580–0.758)	0.663 (0.600–0.722)	0	
DM5	Gradient boosting	0.660 (0.551–0.769)	0.570 (0.424–0.716)	0.654 (0.558–0.750)	0.638 (0.571–0.700)	0	
DM5	Hist. gradient boosting	0.670 (0.571–0.770)	0.603 (0.464–0.742)	0.621 (0.525–0.716)	0.635 (0.571–0.694)	0	
DM5	Neural network	0.680 (0.554–0.805)	0.552 (0.411–0.693)	0.667 (0.588–0.746)	0.644 (0.573–0.709)	15	
BONE	Penalised LR		0.610 (0.473–0.748)		0.576 (0.507–0.641)	0	0.565 (0.488–0.642)
BONE	SVM (RBF)		0.578 (0.428–0.728)		0.563 (0.493–0.631)	0	0.559 (0.480–0.638)
BONE	Random forest		0.565 (0.427–0.703)		0.560 (0.493–0.624)	0	0.558 (0.483–0.632)
BONE	Gradient boosting		0.603 (0.454–0.752)		0.573 (0.506–0.638)	0	0.566 (0.493–0.640)
BONE	Hist. gradient boosting		0.585 (0.432–0.738)		0.552 (0.487–0.616)	0	0.545 (0.473–0.617)
BONE	Neural network		0.606 (0.472–0.740)		0.562 (0.493–0.629)	0	0.548 (0.468–0.627)

Table S4. Paired DeLong comparisons with penalised logistic regression within each held-out cohort (strict L1), Holm-adjusted within endpoint.

Endpoint	Model	Cohort	AUC_LR_minus_model	p	p_Holm
DM5	Random forest	GSE7390	-0.054	0.096	1.000
DM5	SVM (RBF)	GSE2603	0.069	0.124	1.000
DM5	Neural network	GSE7390	-0.052	0.214	1.000
DM5	Gradient boosting	GSE7390	-0.039	0.297	1.000
DM5	SVM (RBF)	GSE7390	-0.020	0.396	1.000
DM5	Random forest	GSE11121	-0.025	0.409	1.000
DM5	Neural network	GSE2603	0.066	0.485	1.000
DM5	Gradient boosting	GSE2603	0.048	0.488	1.000
DM5	Gradient boosting	GSE11121	0.011	0.782	1.000
DM5	Hist. gradient boosting	GSE2603	0.016	0.812	1.000
DM5	Random forest	GSE2603	0.014	0.816	1.000
DM5	Hist. gradient boosting	GSE7390	-0.006	0.866	1.000
DM5	Neural network	GSE11121	-0.009	0.870	1.000
DM5	SVM (RBF)	GSE11121	-0.001	0.980	1.000
DM5	Hist. gradient boosting	GSE11121	0.000	0.995	1.000
BONE	Random forest	GSE2603	0.045	0.242	1.000
BONE	SVM (RBF)	GSE2603	0.033	0.467	1.000
BONE	Hist. gradient boosting	GSE2034	0.020	0.474	1.000
BONE	Hist. gradient boosting	GSE2603	0.025	0.625	1.000
BONE	Random forest	GSE2034	0.007	0.629	1.000
BONE	SVM (RBF)	GSE2034	0.006	0.662	1.000
BONE	Neural network	GSE2034	0.018	0.793	1.000
BONE	Gradient boosting	GSE2603	0.007	0.888	1.000
BONE	Gradient boosting	GSE2034	-0.001	0.951	1.000
BONE	Neural network	GSE2603	0.004	0.962	1.000

Table S5. External calibration (strict L1) per held-out cohort: slope and calibration-in-the-large intercept. Class weighting shifts predicted probabilities upwards, which contributes to the negative intercepts of weighted models.

Endpoint	Model	Cohort	Slope	Intercept
DM5	Penalised LR	GSE11121	0.369	-1.354
DM5	Penalised LR	GSE2603	0.321	-1.478
DM5	Penalised LR	GSE7390	0.228	-2.187
DM5	SVM (RBF)	GSE11121	0.543	-0.213
DM5	SVM (RBF)	GSE2603	0.311	0.667
DM5	SVM (RBF)	GSE7390	0.485	-0.381
DM5	Random forest	GSE11121	0.613	-0.312
DM5	Random forest	GSE2603	0.533	0.412
DM5	Random forest	GSE7390	0.637	-0.675
DM5	Gradient boosting	GSE11121	0.175	0.077
DM5	Gradient boosting	GSE2603	0.123	1.473
DM5	Gradient boosting	GSE7390	0.204	-0.480
DM5	Hist. gradient boosting	GSE11121	0.264	-0.126
DM5	Hist. gradient boosting	GSE2603	0.155	0.971
DM5	Hist. gradient boosting	GSE7390	0.181	-0.396
DM5	Neural network	GSE11121	0.691	-0.411
DM5	Neural network	GSE2603	0.368	0.098
DM5	Neural network	GSE7390	0.767	-0.572
BONE	Penalised LR	GSE2034	0.110	-1.132
BONE	Penalised LR	GSE2603	0.160	-0.859
BONE	SVM (RBF)	GSE2034	0.138	0.216
BONE	SVM (RBF)	GSE2603	0.084	0.101

BONE	Random forest	GSE2034	0.227	-0.306
BONE	Random forest	GSE2603	0.135	-0.081
BONE	Gradient boosting	GSE2034	0.039	2.250
BONE	Gradient boosting	GSE2603	0.130	0.505
BONE	Hist. gradient boosting	GSE2034	0.056	-0.424
BONE	Hist. gradient boosting	GSE2603	0.137	-0.125
BONE	Neural network	GSE2034	0.531	-0.813
BONE	Neural network	GSE2603	0.436	-1.227

Table S6. Decision-curve analysis on pooled strict-L1 predictions (threshold grid 0.05–0.60, step 0.025).

Endpoint	Prevalence	Model	Thresholds_where_net_benefit_exceeds_treat_all_and_none
DM5	0.195	Penalised LR	0.075, 0.100, 0.125, 0.150, 0.175, 0.200, 0.225
DM5	0.195	SVM (RBF)	0.100, 0.125, 0.150, 0.175, 0.200, 0.225, 0.250, 0.275, 0.300
DM5	0.195	Random forest	0.075, 0.100, 0.125, 0.150, 0.175, 0.200, 0.225, 0.250, 0.275
DM5	0.195	Gradient boosting	0.175, 0.200, 0.225, 0.250, 0.275, 0.300
DM5	0.195	Hist. gradient boosting	0.175, 0.200, 0.225, 0.250
DM5	0.195	Neural network	0.100, 0.125, 0.150, 0.175, 0.200, 0.225, 0.250, 0.275, 0.300
BONE	0.226	Penalised LR	0.175, 0.200, 0.225, 0.250
BONE	0.226	SVM (RBF)	0.200, 0.225, 0.250, 0.275
BONE	0.226	Random forest	0.050, 0.100, 0.175, 0.200, 0.225, 0.250
BONE	0.226	Gradient boosting	none
BONE	0.226	Hist. gradient boosting	0.225, 0.250
BONE	0.226	Neural network	0.175, 0.200, 0.225

Table S7. Baselines under strict L1 (mean of cohort AUCs; proliferation score = mean z of BIRC5, CCNB1, CDC20, CEP55, NDC80, MKI67, PTTG1, RRM2, TYMS, UBE2C; NUF2 is absent from HG-U133A; 60 random sets fitted with penalised LR) and signature stability over 50 stratified 80% subsamples without replacement.

Endpoint	Proliferation_score	Random_50_gene_sets	Learned_LR	Jaccard_mean	Jaccard_SD	Distinct_genes	Genes_in_80 pct	Genes_in_all
DM5	0.654	0.577	0.635	0.383	0.062	230	18	4
BONE	0.475	0.524	0.588	0.222	0.053	384	5	1

Table S8. Clinical model (tumour size and grade) versus transcriptomic models trained and tested on the same complete-case patients (5-yr DM).

Held_out_cohort	n	events	Size_and_grade	Penalised LR	SVM (RBF)	Random forest	Gradient boosting	Hist. gradient boosting	Neural network
GSE11121	182	28	0.723	0.662	0.623	0.692	0.633	0.672	0.661
GSE7390	188	36	0.662	0.583	0.651	0.643	0.624	0.570	0.678

S3. Digital twin

Digitisation repeatability from duplicated day-0 readings: SD = 0.0038 (log scale). Pooled baseline residual SD sigma = 0.1202 (11 df). Noise variance of the observed kill rate including the shared day-0 scan: 0.000995 (previous approximation 0.000835).

Forecast-error variance. With OLS slope weights w_t over $t = (-30, -15, 0)$, measurement errors $e_t \sim N(0, \sigma^2)$, prior $k1 \sim N(\mu, \tau^2)$, late change $d = k2 - k1 \sim N(\delta, \sigma_d^2)$ and shrinkage weight K, the error $y12 - \hat{y}12$ equals $e12 - (1 + K)e6 + [K - 6(1 - K)w0]e0 - 6(1 - K)(w_{-30}\, e_{-30} + w_{-15}\, e_{-15}) - 6(1 - K)(k1 - \mu) - 6(d - \delta)$, so $\mathrm{Var} = \sigma^2[1 + (1 + K)^2 + (K - 6(1 - K)w0)^2 + 36(1 - K)^2(w_{-30}^2 + w_{-15}^2)] + 36(1 - K)^2(\tau^2 + \mathrm{Var}\,\hat{\mu}) + 36(\sigma_d^2 + \mathrm{Var}\,\hat{\delta})$. Monte Carlo with 200,000 replicates agreed within 0.4% for K = 0, 0.5 and 1.

Table S9. All models under leave-one-patient-out (L1) and leave-one-lesion-out (L2): MdAPE (%), RMS log error, RMSE (mm³), 95% coverage (%), median hi/lo interval ratio, Winkler interval score and CRPS (log scale). Rate-based models use the analytic predictive variance; LOCF and growth-only use an empirical Gaussian from training residuals.

protocol	model	MdAPE	RMSLE	RMSE	coverage	width_ratio	IS	CRPS
L1	Last observation carried forward	61.905	0.687	232.902	90.909	20.726	3.729	0.393
L1	Growth only (no treatment effect)	75.880	0.728	265.079	90.909	27.167	4.125	0.422
L1	Lesion-only rate (static k)	28.632	0.494	174.579	90.909	8.335	2.556	0.280
L1	Lesion-only rate + drift	27.134	0.449	132.215	100.000	10.204	2.096	0.263
L1	Population rate, no drift	38.537	0.539	146.043	90.909	8.853	3.842	0.304
L1	Population rate + drift	21.254	0.527	108.037	90.909	10.783	3.105	0.293
L1	Twin without drift (Kalman only)	32.442	0.509	136.677	90.909	7.758	3.553	0.281
L1	Digital twin (Kalman + drift)	21.697	0.466	96.920	90.909	9.560	2.821	0.261
L2	Last observation carried forward	61.905	0.687	232.902	90.909	15.818	4.060	0.400
L2	Growth only (no treatment effect)	75.880	0.728	265.079	90.909	18.396	4.293	0.425
L2	Lesion-only rate (static k)	28.632	0.494	174.579	81.818	6.416	3.169	0.282
L2	Lesion-only rate + drift	33.054	0.484	134.757	90.909	7.026	2.084	0.286
L2	Population rate, no drift	26.866	0.519	117.976	90.909	5.920	3.449	0.294
L2	Population rate + drift	39.799	0.504	93.707	90.909	6.508	2.871	0.291
L2	Twin without drift (Kalman only)	30.831	0.487	125.446	81.818	5.681	3.452	0.275
L2	Digital twin (Kalman + drift)	31.369	0.471	97.025	90.909	6.258	2.565	0.275

Table S10. Nested twin ladder: MdAPE with lesion-bootstrap 95% CI. L3 tuned (prior scale, drift weight) = (1, 0.5); L4 variant chosen = twin.

rung	MdAPE	CI_lo	CI_hi	RMSLE
L1	21.697	15.048	49.042	0.466
L2	31.369	11.608	47.525	0.471
L3	25.213	18.603	44.482	0.465
L4	25.213	18.603	44.482	0.465

L5	5.762	2.537	9.722	0.075

Per-fold prior variance under L1: 0.00028, 0.00154 (floor never reached); Kalman gain range L1 0.22–0.61, L2 0.36–0.53. Coverage L1: 10/11 (Clopper–Pearson 58.7–99.8%).

Table S11. Within-patient permutation test (exhaustive derangements within each patient).

Protocol	Twin_MdAPE	Null_median	p	Derangements
L1	21.697	34.732	0.083	11660
L2	31.369	35.063	0.150	11660

Table S12. Paired Wilcoxon tests on absolute log errors (descriptive only: lesions within a patient are not independent).

Comparison	p
L1: twin vs static	1.000
L1: twin vs static_drift	0.765
L1: twin vs pop_drift	0.320
L1: twin vs twin_nodrift	0.953
L1: twin vs locf	0.240
L2: twin vs static	0.765
L2: twin vs static_drift	0.700
L2: twin vs pop_drift	1.000
L2: twin vs twin_nodrift	0.638
L2: twin vs locf	0.147

Table S13. Simulation (300 synthetic cohorts per cell; leave-one-patient-out; statistic = mean absolute log error; 199 within-patient permutations). Scenarios: null (late rate independent of early rate), as estimated from the 11 lesions, strong (residual change halved), strong with measurement SD 0.04. Permutation = % of cohorts with p < 0.05; Wilcoxon = one-sided twin vs population rate.

patients	lesions	scenario	permutation_rejection	wilcoxon_rejection
2	12	null	3.0	4.7
2	12	as_estimated	7.0	13.0
2	12	strong	12.0	16.7
2	12	strong_precise	51.7	38.3
4	20	null	2.3	0.3
4	20	as_estimated	7.3	8.7
4	20	strong	14.0	13.3
4	20	strong_precise	63.3	42.0
10	50	null	3.3	0.3
10	50	as_estimated	14.7	12.7
10	50	strong	24.7	17.7
10	50	strong_precise	97.7	75.0
20	100	null	1.0	0.0
20	100	as_estimated	24.3	12.3
20	100	strong	40.0	22.3
20	100	strong_precise	100.0	95.7
40	200	null	0.3	0.0
40	200	as_estimated	37.0	19.0
40	200	strong	64.7	26.0
40	200	strong_precise	100.0	99.7

Table S14. Illustrative cross-species prior (mouse kmax = 0.493/day, EC50 = 0.225 kV, below the lowest dose; k(30 kV) = 0.490/day = 13.0 × the human median early rate). Forecasts include the population drift; L1 folds.

prior_sd	gain	MdAPE
0.020	0.287	84.569
0.050	0.715	60.577
0.100	0.910	41.304
0.200	0.976	30.883